Computational Drug Discovery in Hepatocellular Carcinoma

An Integrated scRNA-seq and Graph Neural Network Analysis

GEO: GSE166635

3 Notebooks

01

scRNA-seq Analysis

→

02

Target Prioritisation

→

03

GNN Drug Ranking

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OBJECTIVES & DATA

Why this study?

🔬 HCC is molecularly heterogeneous — bulk RNA-seq cannot resolve individual cell populations

🗺️ Map the tumour microenvironment at single-cell resolution

🎯 Prioritise therapeutic targets from dysregulated hub genes

💊 Rank approved drug candidates using graph neural networks

Dataset: GEO GSE166635

10x Genomics Chromium · single-cell RNA sequencing

HCC1

Normal-adjacent liver tissue

16,077

cells

HCC2

Tumour tissue

9,112

cells

25,189 total raw cells · **33,694** genes

Why single-cell?

Bulk RNA-seq averages signals across all cell types, masking the tumour-immune interactions that drive HCC resistance. Single-cell resolution reveals which specific populations are responsible.

METHODS

Computational pipeline

NOTEBOOK 01
scRNA-seq Analysis

QC & Preprocessing
200–2,500 genes/cell · mt < 5%
→ 2,795 cells retained (11.1%)

Normalisation & HVG
10k counts/cell · log1p
2,000 highly variable genes

Clustering
PCA 10 PCs · UMAP · Leiden res=0.5
→ 14 clusters identified

Cell-type Annotation
CellTypist + ScType + SingleR
+ marker scoring → majority vote

DEA
Wilcoxon · padj<0.05 · $|\log_2 FC|>1$
→ 1,385 significant DEGs

GSEA
clusterProfiler (R)
GO-BP/MF/CC · KEGG

NOTEBOOK 02
Target Prioritisation

STRING PPI Network
1,162 DEGs queried
confidence ≥ 400 (medium)

Hub Gene Ranking (high connectivity)
degree + betweenness + closeness
→ composite hub score

Drug–Gene Queries
DGIdb + curated fallback
→ 8,027 interactions

Composite Scoring
5-factor weighted score:
interaction · pubs · phase · approval · hub

NOTEBOOK 03
GNN Drug Ranking

Bipartite Graph
547 gene nodes · 4,978 drug nodes
8,027 edges · 17 features

Three Architectures
GCN · GAT (4 heads) · GraphSAGE
hidden=128 · embed=64 · dropout=0.3

Training
lr=0.005 · 300 epochs
early stopping patience=40

Drug Re-ranking
Best model re-scores all 4,978
candidates by graph topology

RESULTS 01

Quality Control & Tumour Microenvironment

25,189
raw cells
HCC1: 16,077 · HCC2: 9,112

2,795
after QC
11.1% passing filters

14
clusters
Leiden resolution 0.50

1,385
DEGs
335↑ · 1,050↓ in tumour

HCC2 retention: 7.4% vs HCC1: 13.2%

Tumour cells are more metabolically stressed and damaged during tissue dissociation → higher mitochondrial content → correctly excluded by the 5% filter.

Key insight: Macrophages dominate 4 of 14 clusters — tumour-associated macrophages (TAMs) are the central immunosuppressive force in this HCC microenvironment.

Cell populations identified

Macrophages	0, 3, 6, 7	Immunosuppressive
Effector T cells	2, 11	Adaptive anti-tumour immunity
Cytotoxic T cells	8, 12	Direct tumour cell killing
Monocytes	4	Innate immune; TAM precursors
Hepatocytes	1, 9	Liver parenchyma
Plasma / B cells	5, 10	Humoral immunity
Epithelial cells	13	Hepatocytes + cholangiocytes

Differentially Expressed Genes

1,385 significant DEGs · **335** ↑ upregulated in tumour · **1,050** ↓ downregulated in tumour

↑ TOP UPREGULATED IN TUMOUR (HCC2)	
Gene	**Function**
XIST	X-chr. regulator
OLFM4	Stem cell marker
TFF2	Mucosal repair, epithelial protection
CEACAM6	Promotes invasion
MIA	Immune evasion
CTSE	Lysosomal protease

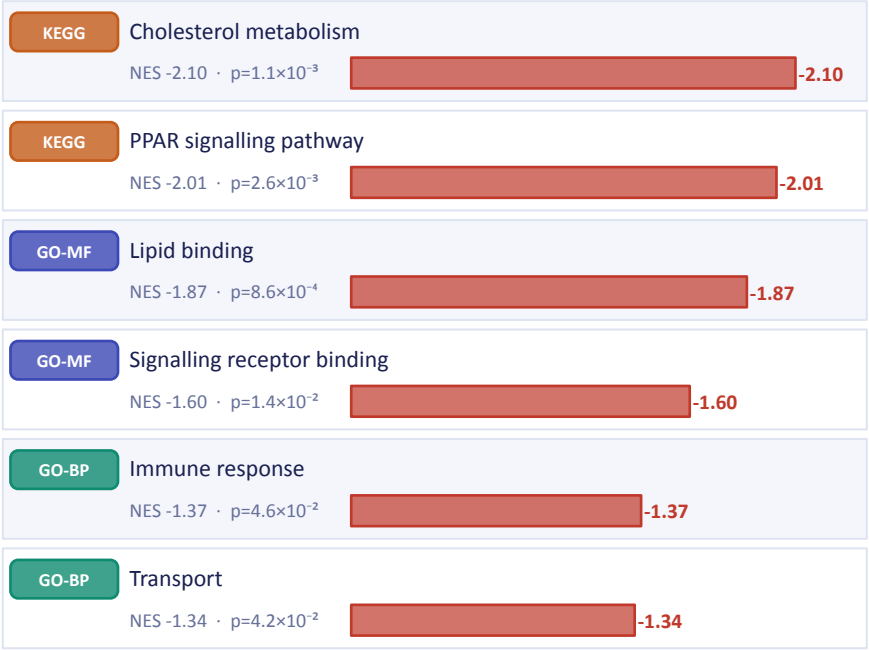
↓ TOP DOWNREGULATED IN TUMOUR (HCC2)	
Gene	**Function**
APOA2	Lipoprotein; normal hepatocyte
ALB	Albumin; liver function
FABP1	Fatty acid binding; hepatocyte
AHSG	Liver-synthesised glycoprotein
AFP	Oncofetal marker; dedifferentiation
TF	Transferrin; iron transport

- Loss of ALB, APOA2, FABP1, and TF indicates loss of hepatocyte differentiation and function.
- Upregulation of OLFM4 (and genes like CEACAM6/MIA) suggests a stem-like, more invasive tumor phenotype,
- XIST reflects epigenetic dysregulation rather than directly indicating proliferation or invasion.

RESULTS 03

Pathway Enrichment

GSEA — all pathways downregulated in tumour



Negative NES (Normalized Enrichment Score) ⟶ pathway is downregulated in tumor

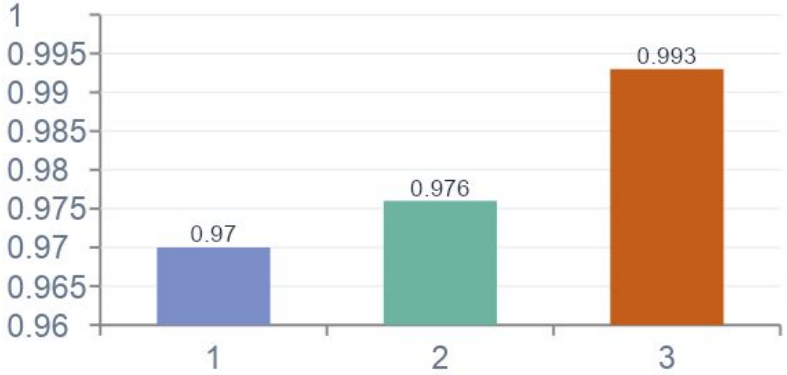
- Loss of hepatocyte metabolic and synthetic functions
- Suppression of lipid and cholesterol pathways
- Reduced transport and signaling capacity
- Weak/uncertain changes in immune activity

RESULTS 04

GNN Performance & Top Drug Candidates

Model comparison (held-out test set)

Model	R^2	MSE	MAE
GCN	0.970	0.001	0.001
GAT	0.976	0.001	0.001
★ GraphSAGE	**0.993**	0.000	0.000



Top 10 drug candidates — GraphSAGE ranking

#	Drug	Gene	Score	Axis
1	Minocycline	AIF1	0.492	Immune
2	Regorafenib	TYROBP	0.489	Validated
4	Deferiprone	FTL	0.485	Iron
5	Deferasirox	FTL	0.482	Iron
7	IFN Gamma-1B	IFNGR2	0.477	Immune
10	Deferoxamine	FTL	0.475	Iron
11	Sorafenib	TYROBP	0.470	Validated
16	Regorafenib	PTEN	0.449	Validated
20	Pentamidine	IL1B	0.445	Immune

CONCLUSION

Key findings & next steps

TAM-dominated microenvironment

4 of 14 clusters are macrophages. Hub genes dominated by macrophage markers (CSF1R, TYROBP, AIF1, CD68). TAMs actively suppress the T-cell response that is simultaneously present but functionally impaired.

Hepatocyte dedifferentiation

Extreme downregulation of ALB (−33.2 FC), APOA2, FABP1, AFP. PPAR & cholesterol pathways suppressed. Tumour cells abandon specialised liver functions for a proliferative, stem-like state.

Iron chelation — convergent vulnerability

Three independent iron chelators (Deferiprone, Deferasirox, Deferoxamine) all converge on FTL across both composite scoring and GNN ranking. Iron sequestration is a meaningful HCC dependency.

Pipeline validated by known drugs

Sorafenib (rank 11) and Regorafenib (rank 2, 16) — the two FDA-approved HCC treatments — were recovered purely from data. This strongly validates the novel predictions: Minocycline (AIF1), IFN Gamma-1B (IFNGR2).

Next steps: Experimental validation of Minocycline (AIF1) & Deferiprone (FTL) in HCC models · Spatial transcriptomics for co-localisation · Larger patient cohorts for stratification